4.10. Frobenius determinant

Enumerate the elements of a finite group G as follows: $g_1, g_2, \ldots, g_n$. Introduce n variables indexed with the elements of G :

$$x_{g_1}, x_{g_2}, \ldots, x_{g_n}.$$

Definition 4.10.1. Consider the matrix X_G with entries $a_{ij} = x_{g_i g_j}$. The determinant of X_G is some polynomial of degree n of $x_{g_1}, x_{g_2}, \ldots, x_{g_n}$ that is called **the Frobenius determinant**, or **group determinant**.

The following theorem, discovered by Dedekind and proved by Frobenius, became the starting point for creation of representation theory (see [**Cu**] and Section 4.11).

Theorem 4.10.2.

$$\det X_G = \prod_{j=1}^{r} P_j(\mathbf{x})^{\deg P_j}$$

for some pairwise nonproportional irreducible polynomials $P_j(\mathbf{x})$, where r is the number of conjugacy classes of G.

We will need the following simple lemma.

Lemma 4.10.3. *Let Y be an $n \times n$ matrix with entries y_{ij}. Then $\det Y$ is an irreducible polynomial of $\{y_{ij}\}$.*

Proof. Let $X = t \cdot \mathrm{Id} + \sum_{i=1}^{n} x_i E_{i,i+1}$, where $i+1$ is computed modulo n, and $E_{i,j}$ are the elementary matrices. Then $\det(X) = t^n - (-1)^n x_1 \ldots x_n$, which is obviously irreducible. Hence $\det(Y)$ is irreducible (since it is so when Y is specialized to X, and since irreducible factors of a homogeneous polynomial are homogeneous, so cannot specialize to nonzero constants). $\square$

Now we are ready to proceed to the proof of Theorem 4.10.2.

Proof. Let $V = \mathbb{C}[G]$ be the regular representation of G. Consider the operator-valued polynomial

$$L(\mathbf{x}) = \sum_{g \in G} x_g \rho(g),$$